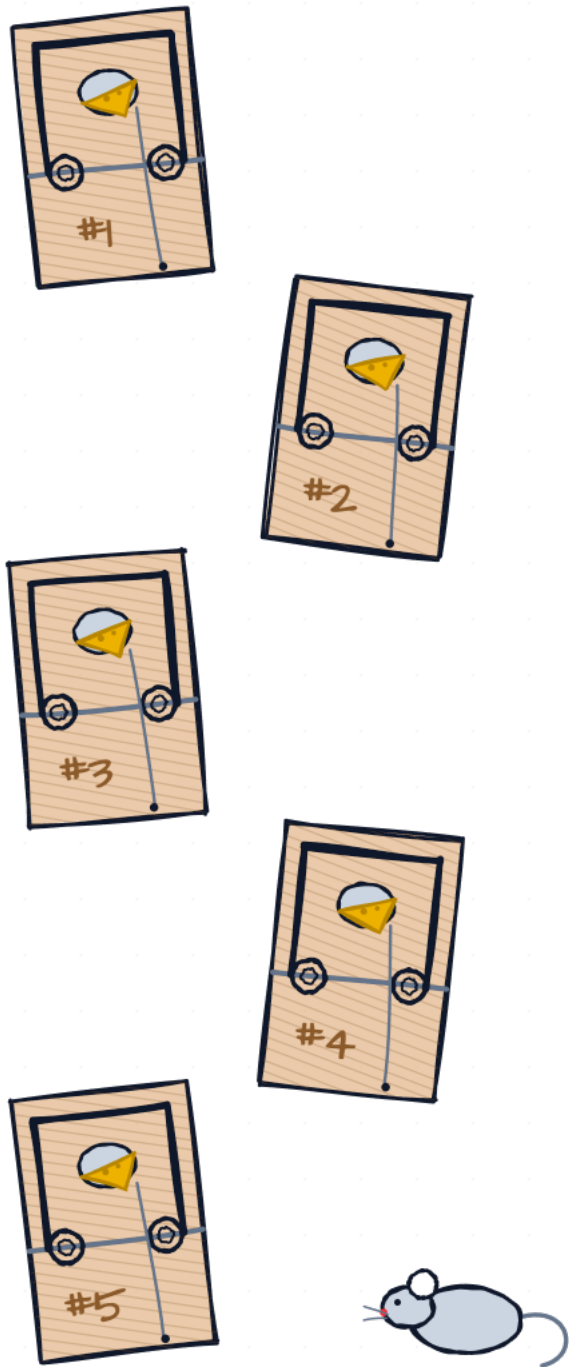




Perri & D'Andrea

Pasticceria
Konditorei
Peter
CAFE

Weges Genuss



Top 5 Spring Boot Testing Mistakes Developers Trap Into

Your tests are green. But do you feel
confident deploying?

Tech Talks South Tyrol #14 · 8th of September 2026

About Philip

- Software Engineer from Erlangen (Bolzano's partner town since 2018), Germany (Bavaria) 🍺
- Blogging & content creation with a focus on testing Java and specifically Spring Boot applications 🌿
- Founder of PragmaTech GmbH - Enabling Developers to Frequently Deliver Software with More Confidence
- Worked together with Martin & Alex from AboutBits last year



Participate During the Talk

Go to menti.com and use the code **XXXX XXXX** to **anonymously** submit answers for the quizzes and add your questions during the talk.

Please answer the **first three questions**:

- Who wrote your tests this week?
- How would you rate your Spring Boot testing knowledge?
- How confident are you deploying on a Friday afternoon (0 to 10)?

Why Test Software?



**In 2026, most of our code is not
written by us.**



Frog told the LLM to write tests for all the code it wrote.

“There,” he said. “Now the model can’t write broken or buggy code.”

“But the model can write broken or buggy tests,” said Toad.

“That is true,” said Frog.

My Overall Northstar for Engineering Excellence

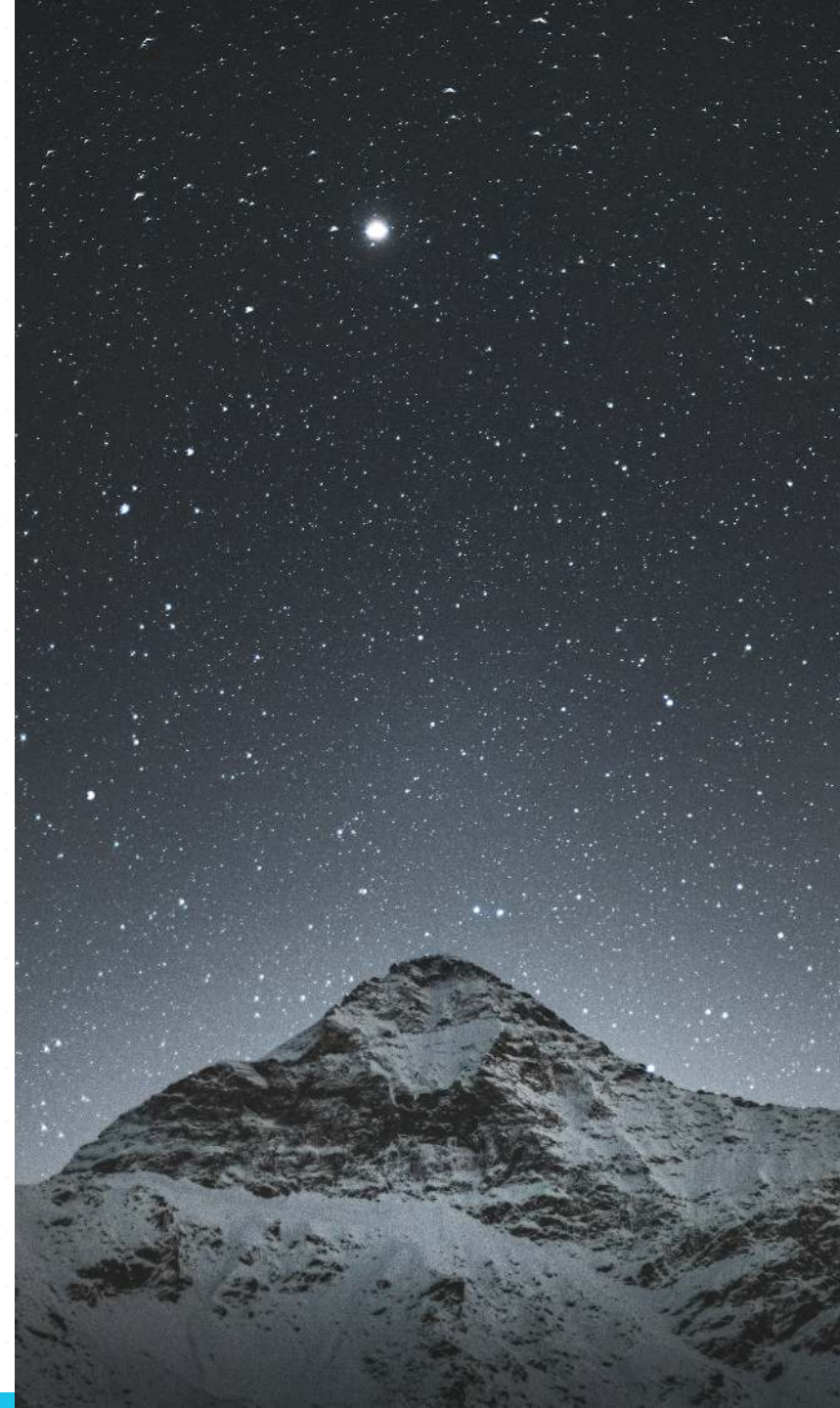
Imagine seeing this pull request on a Friday afternoon:

☐  Update dependency org.springframework.boot:spring-boot-starter-parent to v4.0.0 ✓

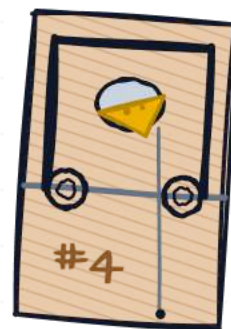
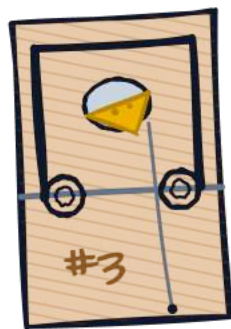
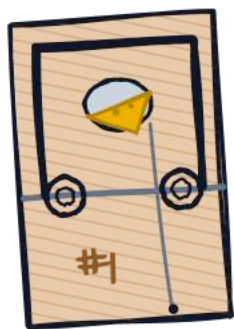
 3

#419 opened on Oct 31 by renovate  · Review required  1 task

How confident are you to merge this major Spring Boot upgrade and deploy it to production once the pipeline turns green?



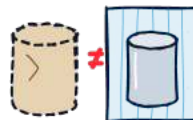
Good tests don't just catch bugs - they give you fast feedback and confident deployments.



Golden Hammer



Context Tax



Production Parity



Phantom Commit



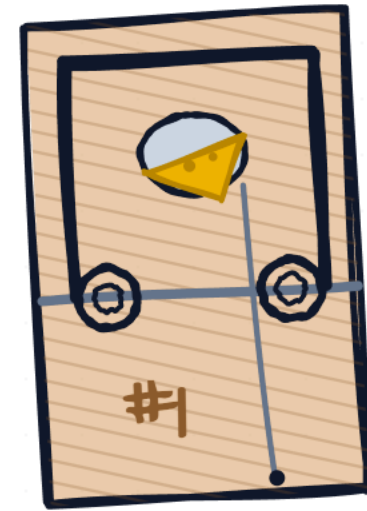
Green Lie



TESTING TRAP #1

The Golden Hammer

If all you have is a hammer, everything looks like a nail.

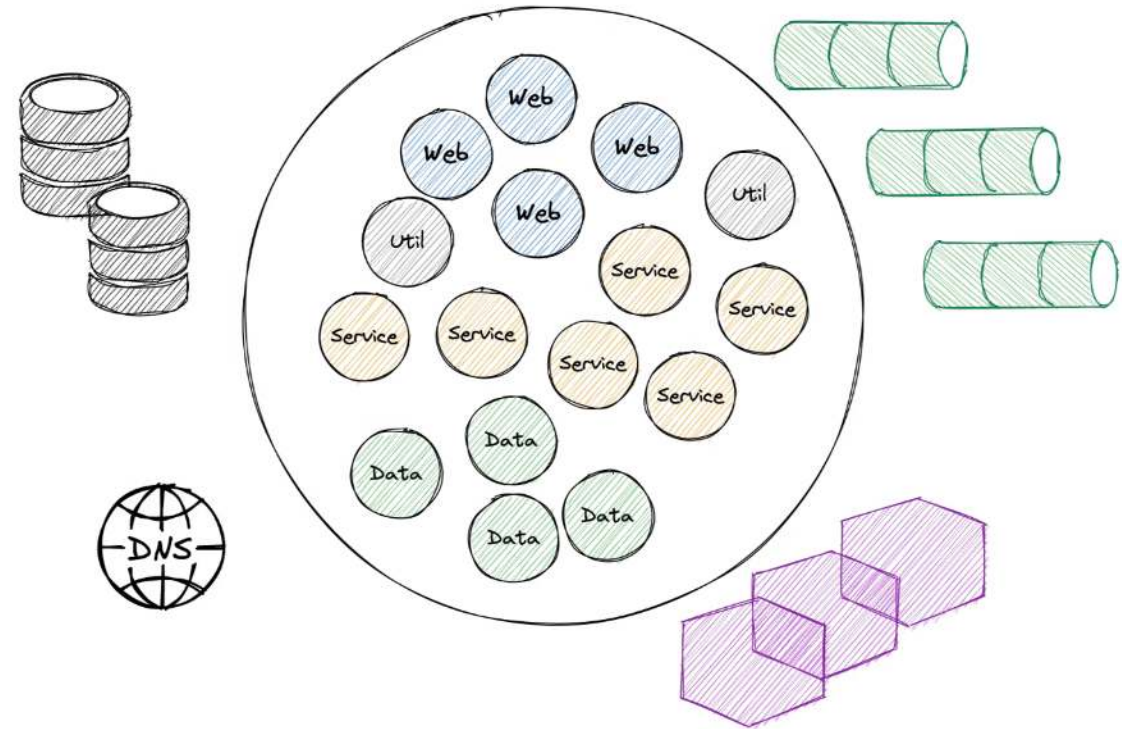


The `@SpringBootTest` Obsession

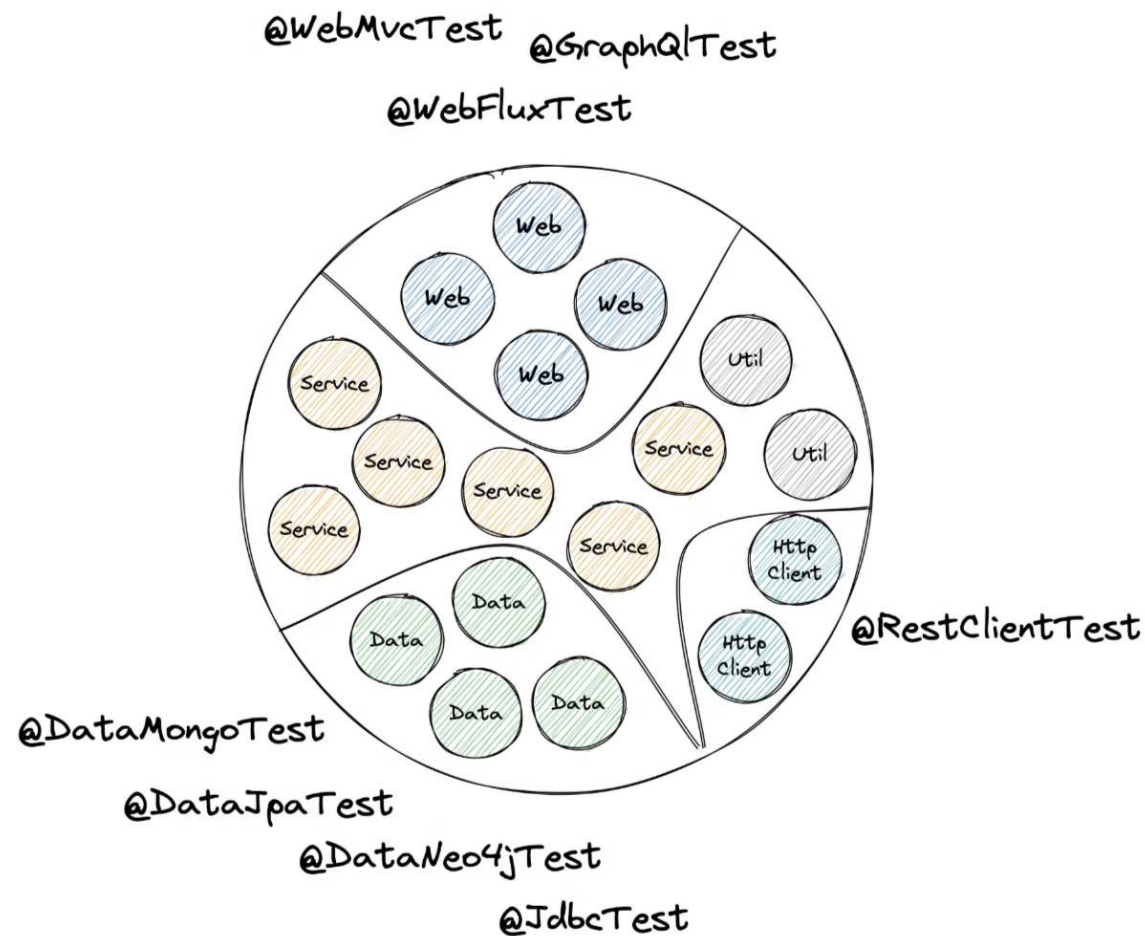
```
@SpringBootTest
class ApplicationTest {

    @Test
    void contextLoads() {
    }

}
```



You Don't Always Need the Entire Context



Decision Paralysis: When to Include a Context

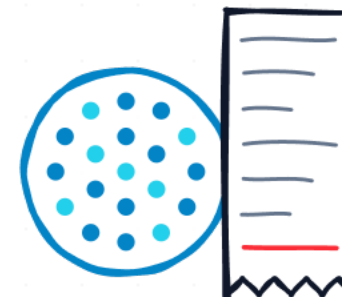
A simplified decision table:

Question	Tool
Does my business logic work?	Plain JUnit 5 + Mockito, no Spring
Does my HTTP layer map, validate, serialize?	<code>@WebMvcTest</code>
Does my query return what I think?	<code>@DataJpaTest</code>
Does my client talk to the remote API?	<code>@RestClientTest</code> + WireMock
Does the whole thing start and work end to end?	<code>@SpringBootTest</code>

TESTING TRAP #2

The Context Tax

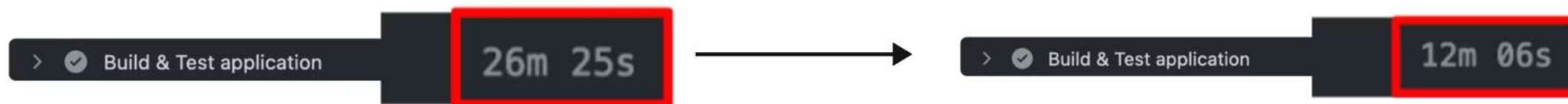
Every `ApplicationContext` you use for testing, comes with a price.



The No. #1 Spring Test Hidden Gem

- Starting the `ApplicationContext` comes with a cost: test execution time
- Every context launch (sliced or full) takes multiple seconds
- Spring Test fixes this with: **TestContext Context Caching**

Results from one of our clients:



Context Caching in a Nutshell

```
1 // DefaultContextCache.java
2 private final Map<MergedContextConfiguration, ApplicationContext> contextMap =
3     Collections.synchronizedMap(new LinkedHashMap<>(32, 0.75f, true));
```

- Spring test builds a unique `ApplicationContext` configuration, by determining which profiles, properties, classes, etc. are included in the test context (stored inside `MergedContextConfiguration`)
- If a subsequent test requires the exact same configuration, Spring simply hands over a "hot" context
- The goal is to reduce the context configuration variety to a bare minimum to have as many cache hits as possible -> fast test execution

How to Identify Context Restarts - Simplified

```
<logger name="org.springframework.test.context" level="TRACE" />
```

```
2022-05-05 14:27:38.136 DEBUG 42500 --- [          main] org.springframework.test.context.cache : Spring test
ApplicationContext cache statistics: [DefaultContextCache@68e62b3b size = 1, maxSize = 32, parentContextCount = 0,
hitCount = 11, missCount = 1]
2022-05-05 14:27:38.136 DEBUG 42500 --- [          main] tractDirtiesContextTestExecutionListener : After test class:
context [DefaultTestContext@325bb9a6 testClass = ApplicationIT, testInstance = [null], testMethod = [null], testException
= [null], mergedContextConfiguration = [WebMergedContextConfiguration@1d12b024 testClass = ApplicationIT, locations =
'{}', classes = '{class de.rieckpil.talks.Application}', contextInitializerClasses = '[]', activeProfiles = '{}',
propertySourceLocations = '{}', propertySourceProperties = '{org.springframework.boot.test.context
.SpringBootTestContextBootstrapper=true, server.port=0}', contextCustomizers = set[org.springframework.boot.test.context
.filter.ExcludeFilterContextCustomizer@5215cd9a, org.springframework.boot.test.json
.DuplicateJsonObjectContextCustomizerFactory$DuplicateJsonObjectContextCustomizer@9257031, org.springframework.boot.test
```

How to Identify Context Restarts - Visualized



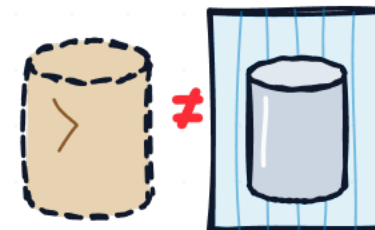
An [open-source Spring Test utility](#) that provides visualization and insights for Spring Test execution, with a focus on Spring context caching statistics.

Overall goal: Identify optimization opportunities in your Spring Test suite to speed up your builds and ship to production faster and with more confidence.

TESTING TRAP #3

The Production Parity Trap

It works on my machine.



Where Tests Drift Away From Production

Every test environment takes shortcuts. Each one is a small drift - and drifts compound:

- **Database:** in-memory H2 instead of the production engine and version
- **Servlet container:** `@SpringBootTest` defaults to a **mocked** servlet environment - `webEnvironment = RANDOM_PORT` starts the real one (Tomcat, etc.)
- **External services:** mocked mail, object storage, message queues, IDPs
- **Schema:** Hibernate `ddl-auto` in tests, Flyway/Liquibase migrations in production
- **Environment:** clock, timezone, locale, JVM flags, OS of the CI runner

The Classic Drift: The In-Memory Database

```
1  @DataJpaTest
2  class OrderRepositoryTest {
3
4      @Autowired private OrderRepository orderRepository;
5
6      @Test
7      void shouldFindOverdueOrders() {
8          // green in two seconds - on an H2 that pretends to be PostgreSQL
9      }
10 }
```

`@DataJpaTest` silently swaps in an embedded database (`replace = Replace.ANY` is the default). H2 in PostgreSQL mode is a **costume, not the engine**.

Disarm It: Close the Gap Step by Step

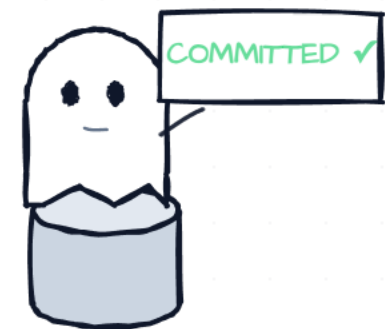
- **Understand your shortcuts**: know where your test environment drifts from production - and decide which drifts you can live with
- **Use Testcontainers**: run the real database, pinned to the production version (`postgres:17.2`) - `@ServiceConnection` wires it up; LocalStack, Kafka, Keycloak, and friends run as containers too
- **Test the app inside Docker**: start the container image you actually ship and run tests against it (JVM flags, base image, OS-level surprises)
- **Canary-test production continuously**: catch what no test environment can - expired SSL certs, rate limits, lost connectivity to cloud resources

TESTING TRAP #4

The Phantom Commit

Schrödingers database commit or

`LazyInitializationException`.



Looks Fine, Right?

```
1  @SpringBootTest
2  @Transactional
3  class CustomerServiceTest {
4
5      @Test
6      void shouldRegisterCustomerAndPublishEvent() {
7          customerService.register(new RegistrationRequest("duke@pragmatech.digital"));
8
9          assertThat(customerRepository.count()).isEqualTo(1);
10         verify(eventPublisher).publishEvent(any(CustomerRegisteredEvent.class));
11     }
12 }
```

What Hibernate Does Behind Your Back

- **Persistence context (first-level cache)**: every entity you save or load lives here - `findById` returns the cached instance, **no SQL executed**
- **Write-behind**: `save()` does not `INSERT` immediately - Hibernate delays SQL until a **flush** (before a query that needs it, or at commit)
- **Flush != commit**: flush sends the SQL, only commit makes it durable, fires the commit-time constraint checks and `AFTER_COMMIT` events
- In a `@Transactional` test the transaction **rolls back**: possibly nothing was ever flushed, and certainly nothing was committed

The Green Test That Proves Nothing

```
1  @Entity
2  public class Customer {
3
4      @Id @GeneratedValue
5      private Long id;
6      private String email;
7
8      public Customer(String email) { this.email = email; }
9      // no no-arg constructor - Hibernate needs it to materialize rows
10 }
```

```
1  @Test
2  @Transactional
3  void greenButMeaningless() {
4      Long id = customerRepository.save(new Customer("duke@pragmatech.digital")).getId();
5      assertThat(customerRepository.findById(id)).isPresent(); // first-level cache, no SELECT
6  }
```

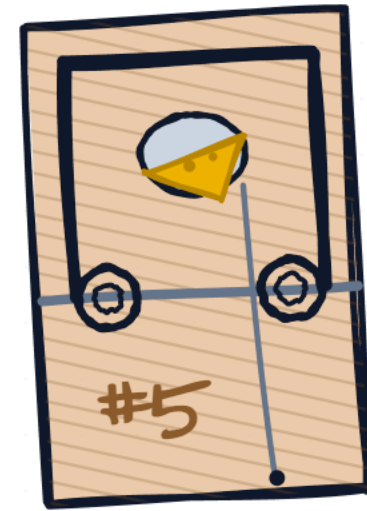
Be Careful on the Transaction Boundary

- The entity is **never materialized from a real row**: the missing no-arg constructor stays invisible until the first production query throws `InstantiationException`
- Rollback at the end of the test: **no commit**, no commit-time constraint checks, no visible change
- `@TransactionalEventListener(phase = AFTER_COMMIT)` **never fires** in this test
- Lazy loading works inside the test transaction and throws `LazyInitializationException` in production
- The gold standard: `@SpringBootTest(webEnvironment = RANDOM_PORT)` + a real HTTP call - the request runs in its **own transaction**, flushes, commits, and reads real rows

TESTING TRAP #5

The Green Lie

Never trust a test you haven't seen failing.



Be Aware of: Watermelon Tests

... green on the outside, red on the inside.

- **100% coverage** and still broken: coverage measures which lines *ran*, not which behavior was *verified*
- A test that asserts the mock proves the mock works
- Auto-generated tests can give you the **feeling** of safety
- Agents produce these at scale: plausible names, green checks, zero judgment

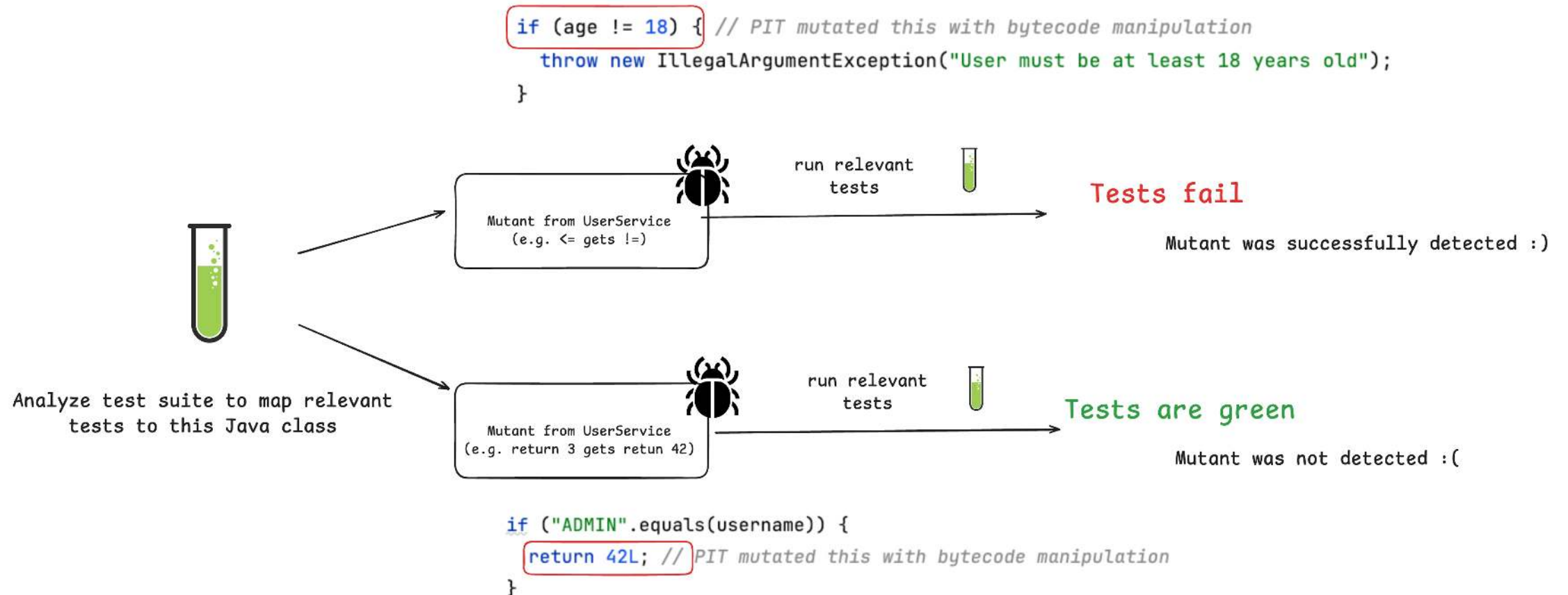
Let's Challenge Code Coverage

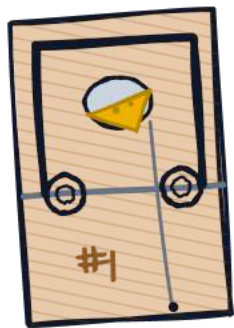
Imagine a set of unit tests for this isolated business logic:

```
1  public Long registerUser(int age, String username) {  
2  
3      if (age <= 18) {  
4          throw new IllegalArgumentException("User must be at least 18 years old");  
5      }  
6  
7      if ("ADMIN".equalsIgnoreCase(username)) {  
8          throw new IllegalArgumentException("Username 'ADMIN' is not allowed");  
9      }  
10  
11     // ...  
12  
13 }
```

Idea: Introduce Regressions to Verify Test Quality

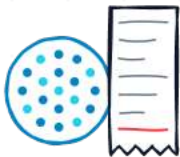
... with the help of PIT:





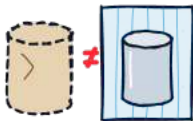
Golden Hammer

✓ test the smallest slice



Context Tax

✓ reuse one cached context



Production Parity

✓ prod-like dependencies



Phantom Commit

✓ test the commit boundary



Green Lie

✓ mutation-test your tests



**AI writes tests in seconds. Trusting
them still takes human judgment.**



What I Do to Build Confidence in the Agentic Coding Era

- **Review generated tests** like a pull request from a new hire - fast, confident, unproven
- **Put the bridges in the repo**: test slices, shared test configuration, Testcontainers, no `@Transactional` on tests
- **Make the rules executable**: ArchUnit rules, my testing conventions in `CLAUDE.md` / `AGENTS.md`, mutation score thresholds (PIT) in CI
- **Keep the feedback loop tight**: fast, trustworthy tests let the agent iterate without me watching every step

My Agentic Testing Setup for Spring Boot

Define a **skillset** for fast & comprehensive tests, including a **test strategy** for the given project:

```
1  spring-boot-testing-skills/  
2  |— unit-testing          fast tests without context bloat  
3  |— slice-testing        right-sized Spring context slices  
4  |— slice-testing-webmvc web layer with @WebMvcTest  
5  |— integration-testing  full-context tests that stay fast  
6  |— testcontainers-setup  real infrastructure, reused containers  
7  |— test-setup-reviewer  flags test anti-patterns for you
```

Each skill ships as `SKILL.md` - readable, editable, yours: pragmatech.digital/agentik-spring-boot-testing-course

Joyful Testing!

Get the slides here:



Reach out any time via [LinkedIn](#) (Philip Riecks) or [Mail](mailto:philip@pragmatech.digital) (philip@pragmatech.digital)

